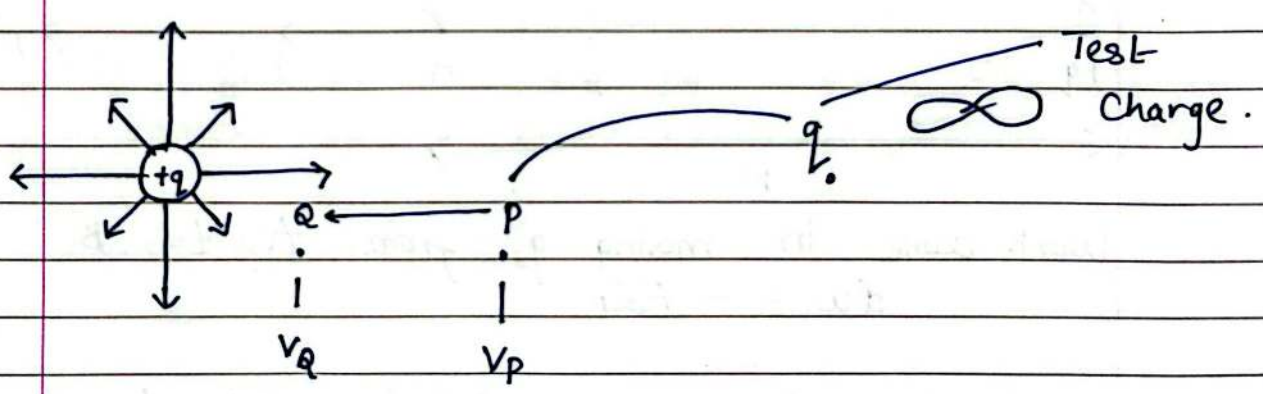


Electrostatic Potential, Energy and Capacitance.

=> Electric Potential

The work done by an external agent in carrying a unit test positive charge from infinity to a point in the electric field without acceleration is called the electric potential at that point.



$$V = \frac{W}{q_0} \quad \text{or} \quad \frac{w}{q}$$

=> $V_Q - V_P$

Electric Potential difference is the work done in bringing a unit test positive charge from a point to another (Here from P to Q)

Dimensions of Electric Potential.

$$\Rightarrow [ML^2 T^{-3} A^{-1}]$$

Unit of Electric Potential

$$\Rightarrow 1V = 1 J/C.$$

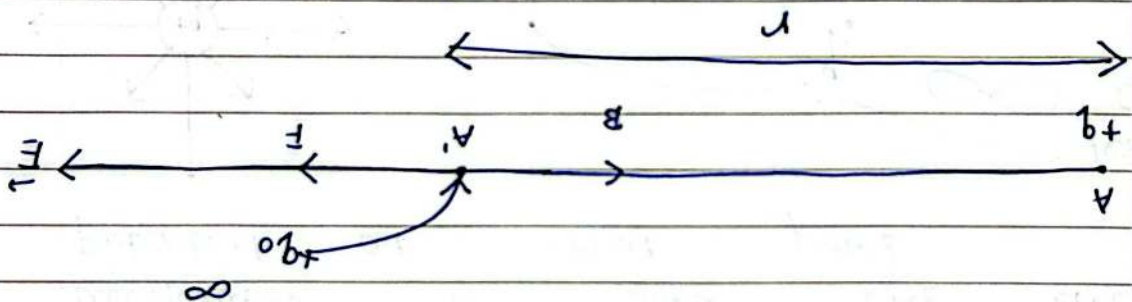
$\Rightarrow$

Electron Volt.

$$1 \text{ Electron Volt} = 1.6 \times 10^{-19} \text{ Joule}$$

$\Rightarrow$

Electric Potential due to a Point Charge



Work done in moving q_0 from A to B

$$dw = -F dr$$

Work done in bringing q_0 from ∞ to A

$$w = \int_{\infty}^A -F dr = - \int_{\infty}^A \frac{kq_0 q_0}{r^2} dr$$

$$w = -kq_0 q_0 \int_{\infty}^A \frac{1}{r^2} dr$$

$$w = -kq_0 q_0 \left[-\frac{1}{r} \right]_{\infty}^A = kq_0 q_0 \left[\frac{1}{\infty} - \frac{1}{A} \right]$$

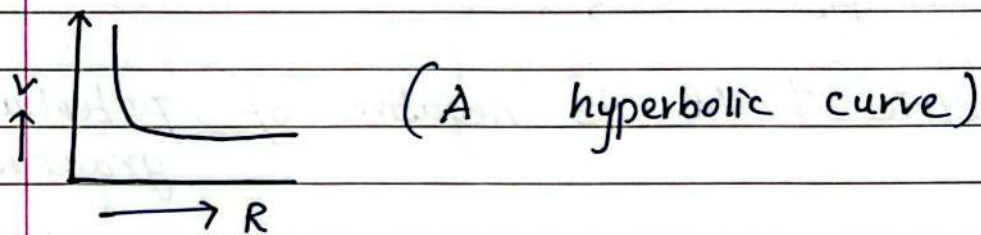
$$w = -\frac{kq_0 q_0}{r}$$

Electric Potential at A

$$V = \frac{W}{q_0} = \frac{kq q_0}{r q_0} = \boxed{\frac{kq}{r}}$$

Here; $V \propto \frac{1}{r}$ (inversely proportional)

Relation with V and R Graph.



$\Rightarrow$ Electric Potential due to a Group of pt charge.

Let $q_1, q_2, -q_3, -q_4$ be at a distance of r_1, r_2, r_3, r_4 from a point.

Potential at that Point is

$$V = \frac{kq_1}{r_1^2} + \frac{kq_2}{r_2^2} - \frac{kq_3}{r_3^2} - \frac{kq_4}{r_4^2}$$

For n number of Charges

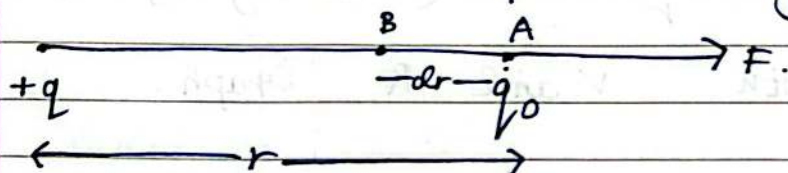
$$V = k \sum_{i=1}^n \frac{q_i}{r_i}$$

⇒ Potential Gradient



Relation between E and V

The rate of change of potential with distance in the electric field is called the potential gradient.



(Electric field is negative of potential gradient).

⇒ Work done in bringing test charge from A to B.

$$dW = -Fdr.$$

Let potential at B be V .

Let potential at A = $(V - dV)$

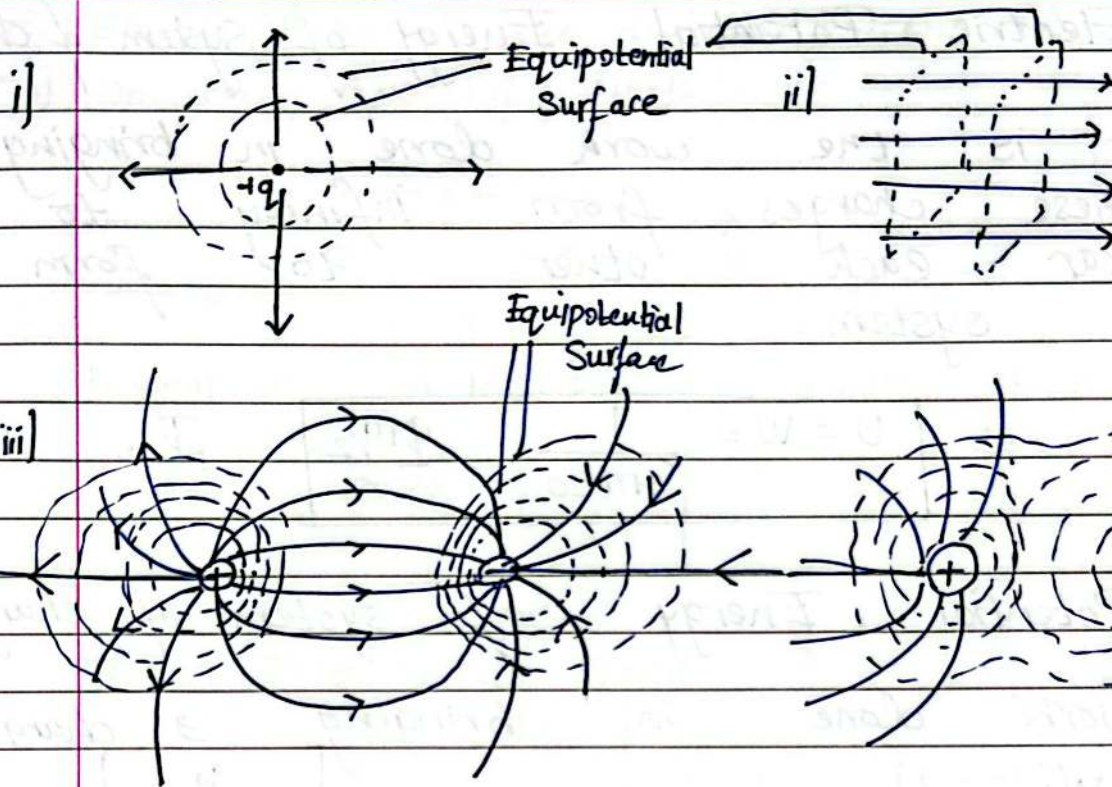
$$[V - (V - dV)]q_0 = q_0 E dr = 0.$$

$$dV = -E dr.$$

$$\boxed{E = -\frac{dV}{dr}}$$

⇒ Equipotential Surface

Any surface over which the electric potential is same everywhere is called an equipotential surface.



Properties of Equipotential Surface.

- i) No work is done in moving a charge between any two points on an equipotential surface.
- ii) The Electric field is always perpendicular to equipotential surface.
- iii) Where equipotential surfaces are closer the electric field is stronger, where they are farther it is weaker.
- iv) Equipotential Surfaces can never intersect each other.

$\Rightarrow$ Electric Potential Energy of System of charges

It is the work done in bringing these charges from infinity to near each other to form a system.

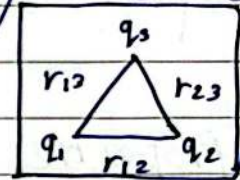
$$U = W = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \text{ J.}$$

$\Rightarrow$ Potential Energy of system of charges

work done in bringing 3 charges

$$W_1 = 0$$

$$W_2 = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r_{12}}$$

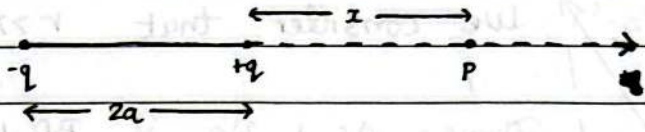


$$W_3 = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \times \frac{q_2 q_3}{r_{23}}$$

$$\text{Total W.D} = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right]$$

⇒ Electric Potential due to an Electric Dipole

* i) On the Axis of Dipole :



Potential at P due to +q

$$V_{+q} = \frac{kq}{(x-a)}$$

Potential at a point due to -q

$$V_{-q} = \frac{k(-q)}{(x+a)}$$

Total Potential at Point P.

$$V_{net} = kq \left[\frac{1}{x-a} - \frac{1}{x+a} \right]$$

$$V_{net} = kq \left[\frac{x+a - x-a}{x^2 - a^2} \right]$$

$$V_{net} = \frac{kq(2a)}{x^2 - a^2}$$

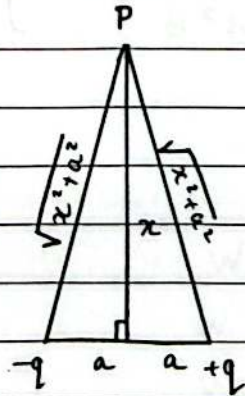
$$V_{net} = \frac{kp}{(x^2 - a^2)}$$

$$p = q \times 2a$$

If $x \gg a$

$$V = \frac{kp}{x^2}$$

* ⇒ At a point P on equator of Dipole.



V due to charge +q

$$= \frac{kq}{(x^2 + a^2)}$$

V due to charge -q

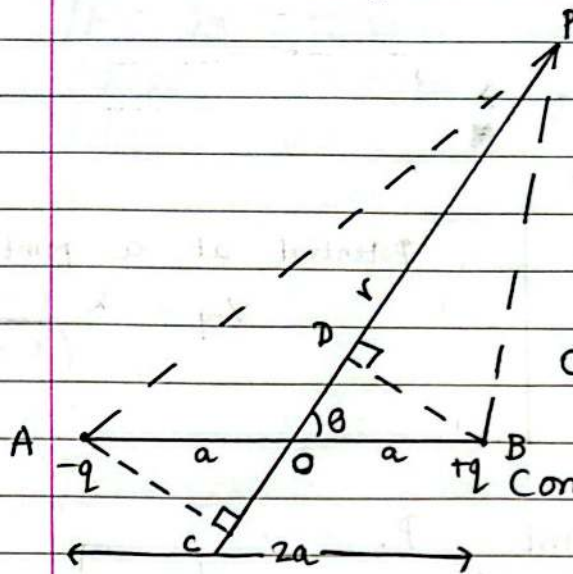
$$= - \frac{kq}{(x^2 + a^2)}$$

Total Potential = $V_1 + V_2 = 0$.

Total Potential at any point on eq. pt is zero.

⇒ Electric Potential at Any other point

* iii] At Any other point:



We consider that $r \gg a$.

Draw $AC \perp PO$, $BD \perp BO$
 Line OP makes angle θ with AB

$$OD = OC = a \cos \theta.$$

as $r \gg a$.

$$\text{Consider: } BP \approx DP = OP - OD \\ = r - a \cos \theta.$$

$$\text{Consider: } AP \approx CP = OP + OC \\ = r + a \cos \theta$$

$$\Rightarrow V_{+q} = \frac{kq}{BP} = \frac{kq}{r - a \cos \theta}$$

$$V_{-q} = \frac{k(-q)}{AP} = \frac{k(-q)}{r + a \cos \theta}$$

$$\Rightarrow \text{Total Potential at Point P} = V_{+q} + V_{-q}.$$

$$= kq \left[\frac{1}{r - a \cos \theta} - \frac{1}{r + a \cos \theta} \right]$$

$$= kq \left[\frac{r + a \cos \theta - r + a \cos \theta}{r^2 - a^2 \cos^2 \theta} \right]$$

$$= kq \left[\frac{2a \cos \theta}{r^2 - a^2 \cos^2 \theta} \right]$$

$$\boxed{V = \frac{kp \cos \theta}{r^2 - a^2 \cos^2 \theta}}$$

Special Cases :

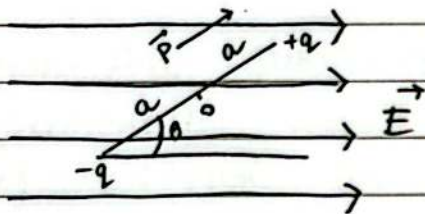
1) If P is an axis of dipole : $\theta = 0^\circ$

$$V = \frac{Kp \cos \theta}{r^2 - a^2 \cos^2 \theta} = \boxed{\frac{Kp}{r^2 - a^2}}$$

2) If P is on equator of dipole : $\theta = 90^\circ$

$$V = \frac{Kp \times 0}{r^2 - a^2 \times 0} = \boxed{0 \text{ (zero)}}$$

⇒ Work done in Rotating an Electric Dipole in Uniform E. F.



Let an Electric Dipole be rotated through $(d\theta)$
 $\tau = P \cdot E \cdot \sin \theta$
 (θ is b/w $\vec{P}$ and $\vec{E}$)

Work done in rotating dipole $d\theta$,

$$dW = \tau \times d\theta.$$

$$dW = P \cdot E \cdot \sin \theta \times d\theta.$$

Total work done in rotating dipole from $\theta_1 - \theta_2$

$$W = \int_{\theta_1}^{\theta_2} dW$$

$$W = \int_{\theta_1}^{\theta_2} P \cdot E \sin \theta d\theta$$

$$W = P \cdot E \int_{\theta_1}^{\theta_2} \sin \theta d\theta.$$

$$W = P \cdot E [-\cos \theta]_{\theta_1}^{\theta_2}$$

$$W = -P \cdot E [\cos \theta_2 - \cos \theta_1]$$

$$U = \boxed{W = P \cdot E [\cos \theta_1 - \cos \theta_2]}$$

1) Special case : i] $\theta_1 = 90^\circ$ $\theta_2 = \theta$ (perpendicular)

$$U = - P \cdot E \cos \theta$$
 (least energy - stable)

ii] $\theta_1 = 0^\circ$ $\theta_2 = \theta$ (parallel)

$$U = P \cdot E [1 - \cos \theta]$$
 — unstable

— If $\theta_2 = 90^\circ$, $U = P \cdot E$

If $\theta_2 = 180^\circ$, $U = 2 P \cdot E$

iii] If $\theta_1 = 180^\circ$ $\theta_2 = \theta$ (antiparallel)

$$U = - P E [1 + \cos \theta]$$
 — Stable

⇒ DIELECTRICS (Insulators)

It is a non-conducting Material that does not allow the flow of free charges

eg:- Glass, mica, Plastic.

⇒ Polar and non Polar Substances.

1. Polar Substances : these have a permanent electric dipole moment.

eg:- water (H_2O), Ammonia (NH_3).

2. Non-Polar Substances : these do not have a permanent dipole moment.

eg:- oxygen (O_2), Nitrogen (N_2)

* Note: All dielectrics are insulators, but not all insulators are dielectrics.

⇒ Capacitance of a Conductor.

The capacitance of a conductor is a measure of its ability to hold electric charge

$$q \propto V.$$

$$q = CV.$$

$$C = \frac{q}{V}.$$

the capacitance of a conductor is defined as the ratio of charge given to the rise in potential of conductor.

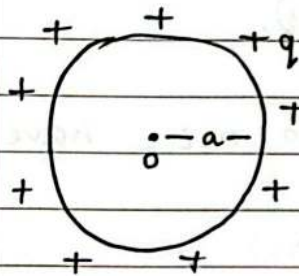
$\Rightarrow$ Factors Affecting Capacitance

- Size of conductor $\propto C$.
- Shape of conductor.
- Separation from other conductors: Smaller distances lead to increasing capacitance.

Unit: farad.

Dimensions: $[M^{-1} L^{-2} T^4 A^2]$.

* $\Rightarrow$ Capacitance of an Isolated Spherical Conductor.



$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{a} \text{ volt.}$$

$$C = q/V.$$

$$C = \frac{q}{q/4\pi\epsilon_0 a.}$$

$$C = 4\pi\epsilon_0 a \text{ farad}$$

Capacitance, of a spherical conductor is directly proportional to its radius.

* $\Rightarrow$ Potential Energy of a Charged Conductor.

$$V = q/C$$

$$dW = V dq = \frac{q}{C} dq$$

$$W = \frac{1}{C} \int_0^q q dq$$

$$= \frac{1}{C} \left[\frac{q^2}{2} \right]_0^q$$

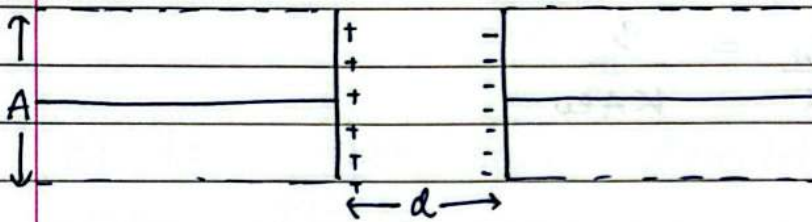
$$\boxed{\frac{1}{2} qV} \text{ or } \boxed{\frac{1}{2} \frac{q^2}{C}} \text{ or } \boxed{\frac{1}{2} C V^2}$$

⇒ CAPACITORS.

A Capacitor is a passive electronic component that temporarily stores and releases electrical energy in the form of an electric field.

⇒ Parallel Plate Capacitors

* two conducting plates of any shape placed parallel to each other.



$$E = \frac{\sigma}{\epsilon_0} \quad \text{or} \quad E = \frac{\sigma}{\epsilon}$$

$$E = \frac{\sigma}{\epsilon_0 K}$$

$$E = \frac{q}{A \epsilon_0 K} \quad \left(\sigma = \frac{q}{A} \right)$$

$$\frac{V}{d} = \frac{q}{A \epsilon_0 K} \quad \left(E = \frac{V}{d} \right)$$

$$\frac{A \epsilon_0 K}{d} = \frac{q}{V} \quad \left(C = \frac{q}{V} \right)$$

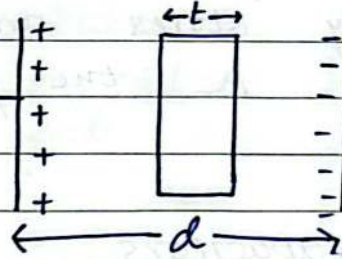
$$\boxed{\frac{A \epsilon_0 K}{d} = C}$$

*⇒ Capacitance is inversely proportional to the distance between the two plates.

↑ ⇒ Capacitance is also directly proportional to the Area of plates

↑ ⇒ also directly proportional to 'K' Medium

* $\Rightarrow$ Capacitance of a parallel plate with a dielectric in between.



$$E_{ext} = \frac{\sigma}{\epsilon_0} = \frac{q}{A\epsilon_0}$$

$$E_{int} = \frac{E_{ext}}{k} = \frac{q}{kA\epsilon_0}$$

Now,

$V = \text{Electric field} \times \text{distance}.$

$$V = E_1 \times t + E(d-t)$$

$$V = \frac{qt}{A\epsilon_0 k} + \frac{q}{A\epsilon_0}(d-t)$$

$$V = \frac{q}{A\epsilon_0} \left[\frac{t}{k} + d-t \right]$$

$$\frac{A\epsilon_0}{\frac{t}{k} + d-t} = \frac{q}{V}$$

$$\boxed{\frac{A\epsilon_0}{d-t + t/k} = C}$$

Special Cases:

a) $d=t$ (fully filled with dielectric)

$$C = \frac{\epsilon_0 A}{d-d+d} = \boxed{\frac{\epsilon_0 K A}{d}}$$

b) Conducting state of thickness t is between plates ($K=\infty$)

$$C = \frac{\epsilon_0 A}{d-t+\frac{t}{\infty}} = \frac{\epsilon_0 A}{d-t}$$

c) It has two dielectric slabs b/w the plates

$$C = \frac{\epsilon_0 A}{d - t_1 - t_2 + \frac{t_1}{K_1} + \frac{t_2}{K_2}}$$

⇒ Energy Stored in a Charged Capacitor.
 [same as Potential Energy]

⇒ Electrostatic Energy Density

$$u = \frac{\text{Energy}}{\text{Volume}}$$

$$u = \frac{1/2 CV^2}{Ad.}$$

$$C = \frac{A\epsilon_0}{d}$$

$$u = \frac{1}{2} \frac{A\epsilon_0}{d} \times \frac{V^2}{Ad.}$$

$$V = Ed$$

$$u = \frac{1}{2} \frac{\epsilon_0 V^2}{d^2}$$

$$u = \frac{1}{2} \epsilon_0 E^2$$

⇒ Force between the plates of a charged parallel plate capacitor.

$$U = \frac{1}{2} QV$$

$$P = F/A$$

$$f d = \frac{1}{2} Q E d.$$

$$P = \frac{1}{2} \frac{Q^2}{A\epsilon_0 A}$$

$$f = \frac{1}{2} QE$$

$$P = \frac{1}{2} \frac{Q^2}{A^2 \epsilon_0}$$

$$f = \frac{1}{2} \frac{Q}{\epsilon_0} \times Q.$$

$$P = \frac{1}{2} \frac{Q^2}{\epsilon_0 A}$$

$$f = \frac{1}{2} \frac{Q^2}{A \epsilon_0}$$

* Combinations

Series : $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} \dots$

Parallel : $C = C_1 + C_2 + C_3 \dots$